Aspandiyar Kurman

Creative Technologist — Audio-reactive Specialist

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Profile

Audio-reactive systems at engine level: custom FFT, onset, tempo, bar and drop detection in C++ and Python drive live visuals in UE5 and TouchDesigner from a file, mic or PC loopback, with Demucs stems and Whisper word timing per layer. Visuals play live or render as frame-exact reels. Code written with AI coding agents under my direction.

Experience

3D Artist — Digiteq Automotive, Virtual Reality Lab — 2023 – now

- Game-asset creation: modelling, texturing, optimisation
- Preparing optimised assets for 3D simulations in VR

Independent real-time & audio-reactive work — Self-directed — ongoing

- Unreal Engine 5 live VJ engine in C++ and HLSL, played with a gamepad
- TouchDesigner GLSL visualizers with a phone web control panel
- Triad: one Python music analyzer drives TouchDesigner live, Houdini simulations and Unreal renders in sync
- Concept, system design, art direction and testing; code written with AI coding agents under my direction

3D Artist (internship) — Cinesco — Jun – Sep 2023

Game-asset design, modelling and sculpting

HMI Front-end Developer — Digiteq Automotive, DTF-D — 2023

HMI skinning; UI installation for Škoda, debugging and testing

Market Researcher — Copy General Technologies — 2022

Research of the Czech market and 3D-print technologies; 3D modelling, consulting, customer service

Printing Operator, junior → senior — Copy General — 2019 – 2022

Typography design services and their sale; order fulfilment, email and phone communication

Selected projects

- **AudioVJ** *(ongoing)* — UE5 VJ engine: custom FFT, beat and bar tracking, drop detection and the singer's pitch drive cameras, lights and materials, live from a file, mic or PC loopback. UE5, C++, HLSL · Case study
- **Perspectiva** — The architecture plays the song: a 16-band equalizer balustrade, a beat-sequencer stair, every scene change on the downbeat, 60 fps. TouchDesigner, GLSL, Demucs · Case study
- **AudioVisualizer** — Tempo tracking, drop and build-up detection and AI-timed lyrics steer 18 shader worlds live, with Kinect, MIDI and phone control. TouchDesigner, GLSL, Python, Whisper · Case study
- **Liquid / Obsidian Bloom** — Four real-time liquids; the ink pours one stratum per beat (5.3 ms mean error) while the singer's pitch steers a glowing stylus. UE5, C++, HLSL, Demucs · Case study

Skills

Audio analysis: FFT, onset detection, tempo tracking with a PLL, bar phase, build-up and drop detection, YIN pitch, Demucs stems, Whisper word timing
Real-time engines: TouchDesigner, Unreal Engine 5, Houdini, Blender, Unity
Code & shaders: GLSL, HLSL, C++, Python (numpy DSP), VEX, JavaScript
Live I/O: OSC, MIDI (MIDI learn, soft takeover), Kinect, NDI, WebSocket, gamepad
Delivery: FFmpeg, NVENC recording, PowerShell
AI-assisted development: directing AI coding agents that write the code while owning concept, system design, testing and review
3D & design: Substance 3D Painter, Quixel Mixer, Photoshop, Figma
Soft skills: teamwork, communication, creativity, management, presentation, organisation

Education

Secondary School of Glassmaking — Kamenický Šenov · glass cutting and engraving; design concepts — 2018 – 2019